

**5G RAN**

# **DTLS Feature Parameter Description**

|              |            |
|--------------|------------|
| <b>Issue</b> | Draft A    |
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# 1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes  
Changes in functions and their corresponding parameters
- Editorial changes  
Improvements or revisions to the documentation

## 1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

### Technical Changes

| Change Description                                                                                | Parameter Change | Base Station Model                                                                                                                                         |
|---------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added support for DTLS by UMPThb series boards. For details, see <a href="#">3.3.3 Hardware</a> . | None             | <ul style="list-style-type: none"><li>• 3900 and 5900 series base stations (macro base stations)</li><li>• DBS3900 LampSite and DBS5900 LampSite</li></ul> |

### Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

---

## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

#### NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

## 2.3 Features in This Document

This document describes the following features.

| RA<br>T | Feature ID  | Feature Name       | Chapter/Section        |
|---------|-------------|--------------------|------------------------|
| NR      | FBFD-010023 | Security Mechanism | <a href="#">3 DTLS</a> |

This feature also provides the public key infrastructure (PKI), PKI redundancy, integrated firewall, and 802.1X-based access control functions to ensure basic network access security. For details about these functions, see *PKI, Equipment Security*, and *Transmission Security Overview*.

## 2.4 Differences

**Table 2-1** Differences between NSA and SA

| Function Name | Difference                      | Chapter/Section        |
|---------------|---------------------------------|------------------------|
| DTLS          | Supported only in SA networking | <a href="#">3 DTLS</a> |

**Table 2-2** Differences between high frequency bands and low frequency bands

| Function Name | Difference | Chapter/Section        |
|---------------|------------|------------------------|
| DTLS          | None       | <a href="#">3 DTLS</a> |

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.



# 3 DTLS

---

## 3.1 Principles

### 3.1.1 Overview

Public IP networks cannot ensure transmission security like traditional time division multiplexing (TDM) networks. Therefore, when operators use public IP networks to carry wireless services, it is recommended that transmission security protocols be used to protect signaling message transmission. The original Transport Layer Security (TLS) protocol is widely used between the application layer and transport layer to ensure network communication security. However, this protocol requires that the data be carried on a reliable transmission channel, indicating that data is transmitted mainly based on the Transmission Control Protocol (TCP). TLS cannot ensure the security of data transmission based on User Datagram Protocol (UDP), which may experience packet loss and reordering. For details about TLS, see *SSL*.

With wide applications of UDP, the Datagram Transport Layer Security (DTLS) protocol is introduced. DTLS is applicable to either reliable protocols such as Stream Control Transmission Protocol (SCTP) or unreliable protocols such as UDP. UDP packet forgery is easy. Therefore, DTLS running over UDP is vulnerable to denial of service (DoS) attacks. DTLS is recommended for SCTP, which defends against forged packets. For details, see section 4.1.2.7 "Handling Invalid Records" in IETF RFC 6347.

The DTLS versions include DTLS1.0 and DTLS1.2. According to "Appendix E.1 General" in 3GPP TS 33.310 V15.0.0, DTLS1.0 is not recommended. Therefore, Huawei base stations support only DTLS1.2. The DTLS function of Huawei base stations supports SCTP, which is currently applicable to the NG-C and Xn-C interfaces.

### 3.1.2 Technical Description

DTLS is used between the application layer and transport layer. It consists of the handshake layer and record layer. For details, see [3.1.2.1 DTLS Protocol Stack](#). The DTLS function of Huawei base stations is implemented as follows: The client and server negotiate the DTLS encryption algorithm suite (referred to as cipher

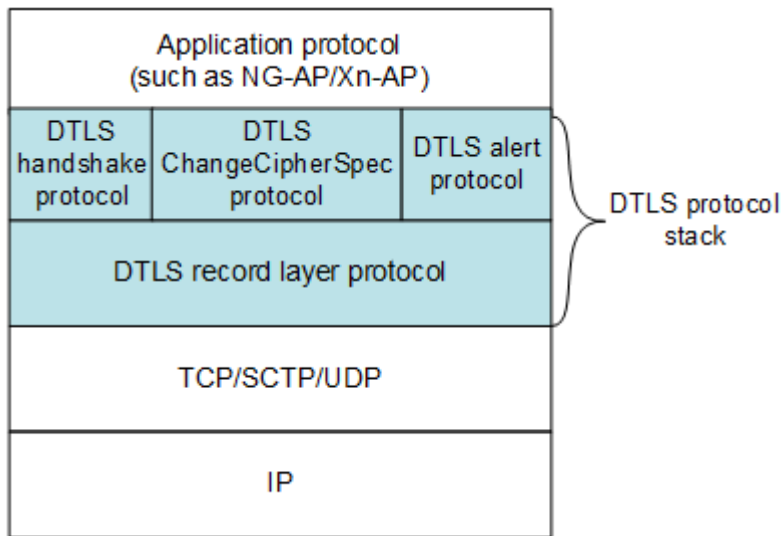
suite hereinafter) to determine the key exchange, identity authentication, symmetric encryption, and message authentication code (MAC) algorithms used in the handshake procedure and subsequent application data transmission procedure. After the negotiation is successful, the client and server continue with the handshake procedure. After the handshake is complete, data transmission between the client and server can be protected by DTLS. For details about the DTLS cipher suite and DTLS connection, see [3.1.2.2 DTLS Cipher Suites](#) and [3.1.2.3 DTLS Connection](#).

3.1.2.1 DTLS Protocol Stack

The DTLS protocol stack involves two layers, as shown in [Figure 3-1](#).

- Handshake layer  
 The handshake layer consists of three protocols:
  - Handshake protocol  
 The handshake protocol is used to set up a security channel between the communicating parties before data transmission begins. During the handshake procedure, the communicating parties authenticate each other, negotiate encryption algorithms, generate keys, and generate initialization vectors.
  - ChangeCipherSpec protocol  
 After the communicating parties agree on a set of new keys, each party sends a ChangeCipherSpec message to notify the peer that subsequent messages will be protected using the newly negotiated keys.
  - Alert protocol  
 An alert message indicates the severity of the alert. When a fatal level alert is generated, the handshake procedure ends immediately.
- Record layer  
 The record layer receives data from or transmits data to the application layer. In addition, the record layer performs security-related operations, such as compression and decompression, encryption and decryption, and MAC calculation.

Figure 3-1 DTLS protocol stack



### 3.1.2.2 DTLS Cipher Suites

For security purposes, Huawei base stations support only the DTLS cipher suites based on perfect forward secrecy (PFS) and authenticated encryption with associated data (AEAD). For details about PFS, see *IPsec*. For details about AEAD, see IETF RFC 5288 and IETF RFC 5289. The cipher suites, as listed in [Table 3-1](#), are specified by the **DTLSPOLICY.CIPHERSUITE** parameter. A smaller sequence number (SN) indicates a higher security level of a DTLS cipher suite and a higher priority during negotiation. Each cipher suite consists of key exchange, identity authentication, symmetric encryption, and MAC algorithms. [Figure 3-2](#) shows an example of the composition of the cipher suite ECDHE\_ECDSA\_GCM256\_SHA384 with SN 1.

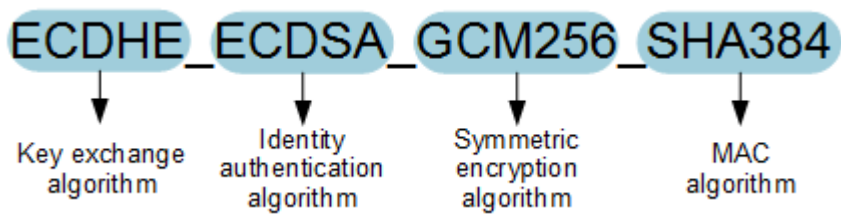
**Table 3-1** DTLS cipher suites

| SN | DTLS Cipher Suite           |
|----|-----------------------------|
| 1  | ECDHE_ECDSA_GCM256_SHA384   |
| 2  | ECDHE_RSA_GCM256_SHA384     |
| 3  | ECDHE_ECDSA_CHACHA20_SHA256 |
| 4  | ECDHE_RSA_CHACHA20_SHA256   |
| 5  | ECDHE_ECDSA_GCM128_SHA256   |
| 6  | ECDHE_RSA_GCM128_SHA256     |
| 7  | DHE_RSA_GCM256_SHA384       |
| 8  | DHE_RSA_CHACHA20_SHA256     |
| 9  | DHE_RSA_GCM128_SHA256       |

 **NOTE**

The preceding cipher suites are sorted based on the algorithm security strength and algorithm performance. A smaller sequence number (SN) indicates higher security and better performance of a DTLS cipher suite, which corresponds to a higher priority. It is recommended that these cipher suites be selected in a descending order of priority based on actual network requirements.

**Figure 3-2** Composition of a DTLS cipher suite



 **NOTE**

The negotiated authentication algorithm type for a cipher suite must be consistent with the certificate type of the base station. If they are inconsistent, the cipher suite cannot be used. For example, a cipher suite cannot be used when the certificate type of the peer base station is RSA, but the cipher suite negotiated between the local and peer base stations connected over an Xn-C interface is ECDHE\_ECDSA\_GCM256\_SHA384, indicating that the authentication algorithm type is ECDSA.

For DTLS1.2, Huawei base stations support only the extension of the uncompressed point format for elliptic curves. Therefore, if an ECDSA certificate uses the point format for its public key elliptic curve, the DTLS connection will be interrupted.

### 3.1.2.3 DTLS Connection

A DTLS connection involves a handshake phase and a data transmission phase. In the handshake phase, the client sends a handshake request to the server to exchange keys, complete identity authentication, and determine the symmetric encryption mode so as to ensure transmission security. If the handshake is successful, the application data is fragmented into manageable blocks that can be encrypted for transmission. This section describes the handshake phase.

The purposes of the handshake are as follows:

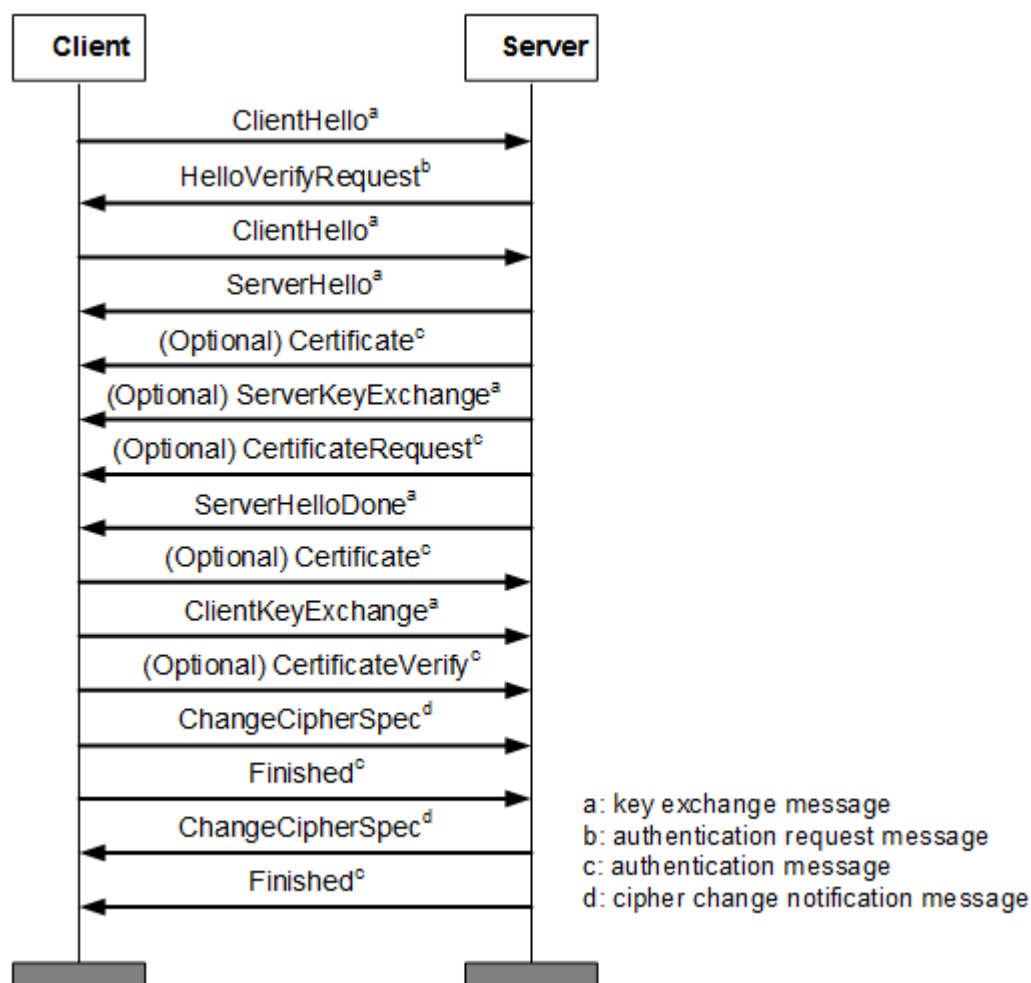
- The client and the server agree on a set of encryption algorithms, MAC algorithms, and algorithm keys to secure data transmission.
- The communicating parties authenticate the identity of each other.

Messages exchanged in the DTLS handshake phase include key exchange messages, authentication request messages, authentication messages, and cipher change notification messages.

- Key exchange messages are used to exchange cipher suites and keys supported by each party.
- Authentication request messages are used to defend against DoS attacks.
- Authentication messages are used to authenticate peer identities through digital certificates.
- Cipher change notification messages are used to instruct the peer to switch to the negotiated algorithm and key.

**Figure 3-3** describes the entire message exchange process between the client and server during a DTLS handshake phase.

**Figure 3-3** Entire message exchange process between the client and server during a handshake



- ClientHello

When a client attempts to connect to a server for the first time, the client sends a ClientHello message to the server. This message contains the protocol versions and a list of cipher suites supported by the client. For details about the list of cipher suites, see [3.1.2.2 DTLS Cipher Suites](#).

After the client receives a HelloVerifyRequest message containing a stateless cookie from the server, the client must send another ClientHello message containing the cookie. Then, the server checks the cookie and continues with the handshake procedure only if it considers the cookie valid.

- HelloVerifyRequest

To defend against DoS attacks, the server responds with a HelloVerifyRequest message after receiving a ClientHello message from the client. This message contains a stateless cookie.

- ServerHello

Upon receiving the ClientHello message, the server checks its cipher suite list as well as that provided by the client in order of priority to determine whether the same cipher suite exists. If there is such a cipher suite, the server selects the cipher suite and sends a ServerHello message to the client. The

ServerHello message contains the protocol version and cipher suite selected by the server. If no, the server sends a handshake failure alarm to the client and ends the handshake procedure.

- Certificate from the server

If the key exchange algorithm in the negotiated cipher suite requires server authentication using a certificate, the server sends a Certificate message containing its certificate to the client. This message is always sent following the ServerHello message. If server authentication is not required, the server will not send a Certificate message.

- ServerKeyExchange

If the key exchange algorithm requires server authentication but the length of the key contained in the Certificate message from the server is insufficient, the server will send a ServerKeyExchange message immediately after the Certificate message to supplement the key information. If the length of the key contained in the Certificate message from the server is sufficient, the server will not send a ServerKeyExchange message.

- CertificateRequest

If the server requests client authentication, the server sends a CertificateRequest message to the client. This message follows the ServerKeyExchange message. If the server does not send a ServerKeyExchange message, the CertificateRequest message follows the Certificate message from the server. If the server does not request client authentication, the server will not send a CertificateRequest message.

- ServerHelloDone

The server sends a ServerHelloDone message to the client to notify the client that the messages indicating support for key exchanges have been sent. After sending this message, the server waits for a response from the client.

- Certificate from the client

When the server requests client authentication, the client sends a Certificate message containing a client certificate to the server after receiving a ServerHelloDone message. If no certificate requested by the server is available, the client sends a Certificate message without a certificate. In this case, the server chooses not to authenticate the client based on the local security policy and continues with the handshake procedure. Alternatively, the server may also respond with a fatal level alert to end the handshake procedure. If the server does not request client authentication, the client will not send a Certificate message.

- ClientKeyExchange

The client sends the server a ClientKeyExchange message containing data used to generate the keys for encryption algorithms and integrity check algorithms. If the client receives a ServerKeyExchange message from the server, the data used to generate the keys is encrypted using the key information carried in the ServerKeyExchange message. If the server requests client authentication, the ClientKeyExchange message must be sent immediately after the Certificate message from the client. Otherwise, the ClientKeyExchange message must be the first message sent by the client after the client receives a ServerHelloDone message.

- CertificateVerify

The CertificateVerify message is used to verify the client certificate. When the client receives a CertificateRequest message from the server, the client sends the server a CertificateVerify message signed using the private key associated with its certificate. If the CertificateVerify message is sent, it must follow the ClientKeyExchange message.

- ChangeCipherSpec

The ChangeCipherSpec message is not a handshake message. However, for timeout and retransmission reasons, the ChangeCipherSpec message must be processed as a message related to the Finished message.

The ChangeCipherSpec message must be received between other handshake messages and the Finished message to instruct the peer to switch to the negotiated algorithm and key. Subsequent messages will be protected using the negotiated algorithm. If no ChangeCipherSpec message is received before the Finished message, a fatal level alert is sent to end the handshake procedure.

- Finished

The Finished message is the first message protected using the negotiated cipher suite. The recipient of the Finished message must verify that the content is correct. Once a party has sent a Finished message and has received and verified the Finished message sent by the peer, the party can start to send and receive application data on transmission links. The Finished message is sent after a ChangeCipherSpec message, indicating that the key exchange and authentication are successful.

After the handshake is successful, the two parties enter the data transmission phase and start to exchange application data based on the cipher suite negotiated through the handshake.

When a base station works as a DTLS client or server, certificates in the DTLS trust group specified by the **SCTPHOST.TRUSTGROUPID** parameter can be used to verify the device certificates of the peer end. If this parameter is not specified, all trust certificates of the base station are used to verify the device certificates of the peer end by default.

 **NOTE**

For details about trust certificates and trust groups, see *PKI*.

### 3.1.3 Application

According to section 9.2 "Security mechanisms for the N2 interface" and section 9.4 "Security mechanisms for the Xn interface" in 3GPP TS 33.501 V15.1.0, Huawei base stations in SA networking support DTLS-based signaling message transmission security over the NG-C and Xn-C interfaces.

 **NOTE**

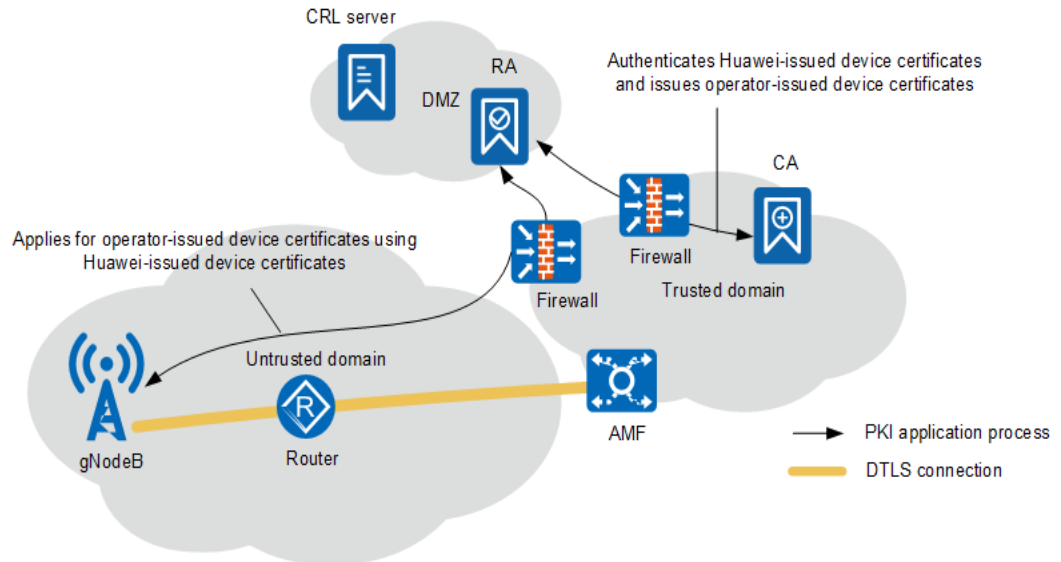
Currently, only the NG-C and Xn-C interfaces support DTLS. When DTLS is activated for other interfaces, ALM-25888 SCTP Link Fault will be reported.

#### 3.1.3.1 NG-C Interface

When DTLS is applied to the NG-C interface, the gNodeB and the access and mobility management function (AMF) authenticate each other using digital

certificates and set up a DTLS connection to protect control-plane transmission. For details about how the base station applies for a certificate from the operator's CA server, see *PKI*. **Figure 3-4** shows the NG-C networking.

**Figure 3-4** DTLS applied to the NG-C interface



In this scenario, the process of setting up a DTLS connection is as follows:

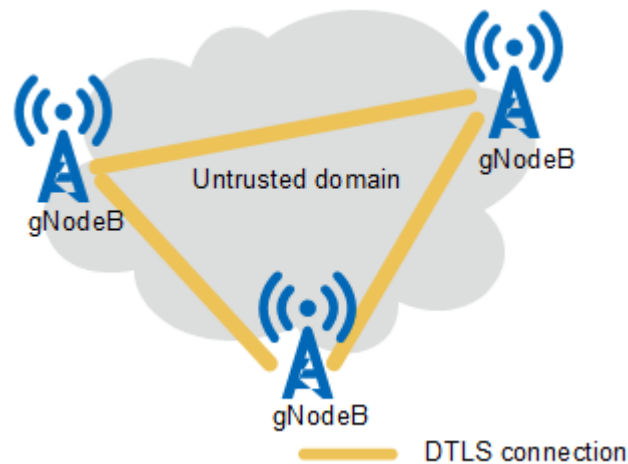
1. The gNodeB sets up an SCTP connection with the AMF.
2. The gNodeB, functioning as a DTLS client, initiates a DTLS handshake with the AMF.
3. During the handshake, the gNodeB negotiates the DTLS version and cipher suite with the AMF and authenticates the AMF digital certificate. The AMF may authenticate the gNodeB, depending on the AMF security policy. After the authentication is successful, DTLS is applied to protect signaling messages on the NG-C interface.

### 3.1.3.2 Xn-C Interface

When DTLS is applied to the Xn-C interface, signaling messages over the Xn-C interface between gNodeBs are protected by DTLS, as shown in **Figure 3-5**. Before link establishment, the operator-issued device certificate and the operator's root certificate need to be obtained from the operator's public key infrastructure (PKI) system. For details, see *PKI*.



**Figure 3-5** DTLS applied to the Xn-C interface



In this scenario, the process of setting up a DTLS connection is as follows:

1. A local gNodeB sets up an SCTP connection with the peer gNodeB.
2. The local gNodeB, functioning as a DTLS client, initiates a DTLS handshake with the peer gNodeB.
3. During the handshake, the local and peer gNodeBs negotiate the DTLS version and cipher suite and perform two-way authentication. After the authentication is successful, DTLS is applied to protect signaling messages on the Xn-C interface.

## 3.2 Network Analysis

### 3.2.1 Benefits

A DTLS connection is set up between a client and a server to ensure effective and secure communications between the two parties. DTLS provides the following security functions:

- Identity verification: The communicating parties authenticate each other before setting up a DTLS connection.
- Confidentiality: Data transmitted between communicating parties is encrypted to prevent eavesdropping.
- Integrity protection: Integrity protection is provided for data transmitted between the communicating parties to prevent data tampering during transmission.

### 3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

After DTLS is enabled, the CPU usage increases.

## 3.3 Requirements

### 3.3.1 Licenses

None

### 3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 3.3.2.1 Prerequisite Functions

| RAT                                                     | Function Name            | Function Switch | Reference                        |
|---------------------------------------------------------|--------------------------|-----------------|----------------------------------|
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | Security Mechanism (PKI) | None            | <i>PKI</i>                       |
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | NG self-management       | None            | <i>NG and Xn Self-Management</i> |
| NR FDD<br>Low-frequency NR TDD<br>High-frequency NR TDD | Xn self-management       | None            | <i>NG and Xn Self-Management</i> |

#### 3.3.2.2 Mutually Exclusive Functions

None

#### 3.3.2.3 Function Impacts

None

### 3.3.3 Hardware

#### Base Station Models

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

#### Boards

Only the UMPTe, UMPTg, UMPTga, UMPTha, and UMPThb series boards support this function.

#### RF Modules

No requirements

### 3.3.4 Others

None

## 3.4 Operation and Maintenance

### 3.4.1 Data Configuration

#### 3.4.1.1 Data Preparation

**Table 3-2** describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

**Table 3-2** Parameters used for activation

| Parameter Name | Parameter ID                   | Setting Notes                                                                       |
|----------------|--------------------------------|-------------------------------------------------------------------------------------|
| DTLS Policy ID | <b>DTLSPOLICY.DTLSPOLICYID</b> | Set this parameter to <b>0</b> .                                                    |
| Cipher Suite   | <b>DTLSPOLICY.CIPHERSUITE</b>  | Only secure cipher suites are supported. By default, all cipher suites are enabled. |

| Parameter Name         | Parameter ID                   | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DTLS Policy ID         | <b>SCTPHOST.DTLSPOLICYID</b>   | It is recommended that this parameter be set to <b>0</b> to enable DTLS. If this parameter is set to <b>NULL</b> , the plaintext SCTP link is set up, which may cause information disclosure or tampering.                                                                                                                                                                                                                                                                      |
| DTLS Connection Policy | <b>SCTPHOST.DTLSCONNPOLICY</b> | Set this parameter based on the network plan. If DTLS is enabled on the entire network, it is recommended that this parameter be set to <b>DTLS_ONLY</b> to improve security. If DTLS is not enabled on the entire network, it is recommended that this parameter be set to <b>DTLS_PRIORITY</b> to prevent service loss. However, in this case, plaintext SCTP links may be used, which may cause information disclosure or tampering.                                         |
| Trust Group ID         | <b>SCTPHOST.TRUSTGROUPID</b>   | This parameter is used for verifying the device certificates of the peer end. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the <b>TRUSTGROUP</b> MO are used to authenticate the peer end. If this parameter is set to <b>65535</b> , no trust group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end. |

**NOTICE**

It is recommended that a unified DTLS enabling policy be adopted on the entire network. If DTLS is enabled for some base stations but is not supported by other base stations on a network, DTLS connection setup will fail, causing service loss. It is recommended that DTLS be enabled after all gNodeBs are upgraded to the current version. To ensure service continuity, DTLS is disabled by default in the current version.

By default, all options of the **SCTPHOST.SCTPAUTHCHUNKLST** parameter are selected. This parameter can be set according to the transport network plan to prevent Sctp link disconnection and service loss. It is recommended that all options of the **SCTPHOST.SCTPAUTHCHUNKLST** parameter be selected to request the peer Sctp entity to authenticate these Sctp blocks to improve link security. If not all options are selected, some Sctp blocks are not authenticated. In this case, Sctp information may be tampered with.

### 3.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

This section describes only the MML command examples for adding a DTLS configuration policy and binding the DTLS configuration policy to an Sctp host. For details about the MML command examples for NG and Xn interface setup, see *NG and Xn Self-Management*.

### Activation Command Examples

```
//Adding a DTLS configuration policy
ADD DTLSPOLICY: DTLSPOLICYID=0,
CIPHERSUITE=ECDHE_ECDSA_GCM256_SHA384-1&ECDHE_RSA_GCM256_SHA384-1&DHE_RSA_GCM256_SH
A384-1&ECDHE_ECDSA_CHACHA20_SHA256-1&ECDHE_RSA_CHACHA20_SHA256-1&DHE_RSA_CHACHA20_
SHA256-1&ECDHE_ECDSA_GCM128_SHA256-1&ECDHE_RSA_GCM128_SHA256-1&DHE_RSA_GCM128_SHA2
56-1;
//Binding the DTLS configuration policy 0 to an Sctp host
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412, SCTPTEMPLATEID=0, DTLSPOLICYID=0,
DTLSCONNPOLICY=DTLS_ONLY, TRUSTGROUPID=65535;
```

### Deactivation Command Examples

```
//Unbinding the DTLS configuration policy from the Sctp host
MOD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="5.5.3.37", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38412, SCTPTEMPLATEID=0, DTLSPOLICYID=NULL;
```

### 3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 3.4.2 Activation Verification

Run the **DSP SCTPLNK** command to query the status of DTLS connections. If the value of the **DTLS Link Status** parameter is **Up** for all DTLS connections, the DTLS function is successfully activated.

## 3.4.3 Network Monitoring

After DTLS is enabled, perform the following operations for function monitoring:

- Method 1:  
Log in to the LMT, and choose **Trace > Common Services > SCTP Trace** to trace DTLS messages.
- Method 2:
  - a. Log in to the MAE, choose the MAE-Access, and then choose **Monitor > Signaling Trace > Signaling Trace Management**.
  - b. On the **Signaling Trace Management** tab page, query the DTLS messages in the SCTP tracing result.

# 4 Parameters

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The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

## **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.

**Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End



# 5 Counters

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The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 6 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 7 Reference Documents

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- 3GPP TS 33.310 V15.0.0: "Network Domain Security (NDS); Authentication Framework (AF)"
- 3GPP TS 33.501 V15.1.0: "Security architecture and procedures for 5G system"
- 3GPP TS 38.104 V17.8.0: "Base Station (BS) radio transmission and reception"
- IETF RFC 6347: "Datagram Transport Layer Security Version 1.2"
- IETF RFC 5288: "AES Galois Counter Mode (GCM) Cipher Suites for TLS"
- IETF RFC 5289: "TLS Elliptic Curve Cipher Suites with SHA-256/384 and AES Galois Counter Mode (GCM)"
- *SSL*
- *PKI*
- *IPsec*
- *NG and Xn Self-Management*
- *SRAN Networking and Evolution Overview*
- *Transmission Security Overview*
- *Equipment Security*